

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and executed test plans for embedded software components of **next-generation Free Space Optics (FSO)** terminals, identifying and documenting defects with a **95% defect detection rate**.
- Automated test case execution for **backend applications** using **Python**, integrating scripts into **CI/CD pipelines** to reduce manual testing effort by **40%**.
- Performed root cause analysis on software defects within mission-critical systems, collaborating with development teams to ensure high-reliability software for **space-grade hardware**.
- Contributed to the testing of **networking protocols** for satellite communication systems, analyzing packet data and verifying protocol compliance against specifications.
- Collaborated within an **Agile** environment, communicating test progress and results directly to developers and product owners, facilitating rapid iteration cycles.
- Assisted in the analysis of technical specifications and designs to define test scopes and strategies for **high-reliability systems** with stringent quality requirements.

SKILLS

Languages – Python, C, C++, JavaScript

Testing Concepts – Software Testing, SDLC, Test Planning, Test Execution, Root Cause Analysis, Hardware-in-the-loop testing

Embedded Systems – Embedded Software, FSO Terminals, Space-grade Hardware

Backend & CI/CD – Backend Applications, CI/CD, Jenkins, BitBucket Pipelines, GitLab CI, Docker

Networking – Networking Protocols

General – Agile, Technical Specifications, Requirements Analysis

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin

02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India

01/2021

PROJECTS

Embedded Systems Test Suite for Satellite Communications

- Designed and implemented automated test scripts using **Python** to verify embedded software functionality for satellite communication modules.
- Executed manual and automated tests against **space-grade hardware** prototypes, meticulously documenting defects and their potential impact on orbital operations.

- Analyzed test results and generated reports for stakeholders, providing insights into software quality and readiness for deployment.

Backend API Integration and Performance Testing

- Developed a suite of automated tests for backend APIs using **Python** and **FastAPI**, focusing on integration points with embedded systems.
- Performed load testing on backend services to identify performance bottlenecks, achieving a **20% improvement** in response times under simulated peak conditions.
- Integrated automated testing into a **Jenkins** CI/CD pipeline, ensuring continuous verification of API stability and functionality.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)